Housing First or Treatment First?

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Response to Referee 3

Major and minor concerns, responses in red

Thank you for the report, and there is no need to apologise for the timing. Your second major concern contains the sharpest theoretical question in any of the four documents, and answering it properly has produced what I now think is the most interesting table in the paper.

One note before the comments. The paper has a new title: it was submitted as *Housing First or Treatment First?* and is now *Housing First, Children Later*. The reason is the substance of the revision rather than a relabelling, and it will be clearest at M2(c), which asks where the externalities come from.

Major concern 1: connection to empirical evidence and stylised facts

M1(a) Present stylised facts prominently

> *"The nature of the model, with its large number of parameters, makes it relatively opaque. I believe it would be helpful for the paper to include a few stylized facts more prominently, notably on the empirically observed correlations between drug use, voucher receipt, and homelessness."*

Done. There is now an empirical section before the model, presenting the joint distribution of drug use, substance use disorder, voucher receipt, employment and homelessness, computed from NSDUH and from the HUD point in time and Housing Inventory files, with the code in the replication package.

The Associate Editor asks for the same thing in his comment A1 and adds a requirement I have adopted: the stylised model that follows should generate these facts, so that the reader can see which features of the environment the facts are pinning down before meeting the full quantitative model.

M1(b) Survival and unemployment probabilities are not transparent

> *"Survival probabilities (especially their dependence on homelessness and drug use) and unemployment probabilities are an essential input into the model's calibration and welfare implications. It would be good to present more prominently some empirical evidence on their correlation with homelessness and drug use, which are currently borrowed from other papers but not always set transparently (for example for unemployment transition probabilities)."*

This comment was better aimed than I realised at the time, and rebuilding these objects from primary sources found two errors that I report here rather than leave to be discovered.

- **The unemployment rate given a substance use disorder had no traceable provenance.** It was set at 26.9 per cent. Measured directly in the 2024 NSDUH microdata for ages 26 and above, weighted, it is 16.25 per cent for moderate or severe drug use disorder. The old

figure implied a relative risk of 5.49 against an overall rate of 4.9 per cent; the data give 2.44. The model's channel from substance use to employment was running roughly 2.25 times too strong.

- **The homelessness mortality multiplier was a single flat number, 1.6.** Baggett et al. (2013) give race adjusted all cause rate ratios of 8.6 for men and 9.6 for women at ages 25 to 44, 4.5 at 45 to 64, and 1.1 at 65 to 84. Both the level and the age profile were wrong, and the multiplier is now an age profile built from that source. The life table has also been replaced with a current vintage.

There is now a dedicated subsection that sources every hazard and every multiplier, with the mortality multipliers reported alongside the standardised mortality ratios they imply, so a reader can check them against Baggett et al. and Degenhardt et al. directly. Transparency here is not a presentational matter: two of these objects were wrong, and they were wrong in ways that mattered for the paper's central claim.

[**Built since the first draft of this letter.**] Built. Baseline survival is the SSA period life table for ages 26 to 85, with the sexes combined by survivor weight rather than averaged equally, since by the mid eighties the two survivor populations differ enough that equal weights are simply wrong. The curve it replaces understated mortality by nearly half at the top of the range, $q = 0.045$ at 85 against an actual 0.081.

The homelessness and addiction multipliers turned out not to be separable and are now derived jointly, because Baggett et al.'s homelessness ratio is measured on a population that mostly also carries a disorder, so netting that out depends on where the disorder multiplier lands. Because the disorder multiplier falls with age, so does the correction, and the homelessness multiplier at 56 comes out at 3.14 rather than the 1.97 a flat calculation gives. The level of the disorder multiplier stays calibrated, for the reason given under M2(d): it is the second stage's only instrument.

M1(c) The income to drug use relationship is not shown

> *"... the relationship between income and drug use (severe or mild) is key to the calibration of b and the income-dependent penalties in the model ... but its empirical counterpart does not feature very prominently in the paper. It would be helpful to get a sense of this joint distribution in the data."*

Plotted from the NSDUH microdata, and this exercise changed a target.

The relationship is not one gradient but two, and the distinction is the paper's mechanism. For ages 26 and above, average income relative to the population average is 0.951 for people who *use* drugs and 0.687 for people with a moderate or severe use *disorder*. Use is close to income neutral; it is *progression* to disorder that concentrates at the bottom.

Both ratios are now calibration targets, which is a sharper test than targeting either alone, and it is the right test for a model whose mechanism is that the expected cost of progression rises with lifetime resources. Referee 1's comment 2(c)(i) compared a model moment over users with a data moment over people with a disorder; separating them was necessary to see what was actually going on.

Major concern 2: modelling of the drug addiction process

M2(a) Reducing addiction mechanically raises consumption utility

> *"... reducing addiction lowers θ_d ... and thus increases directly consumption utility from housing and non-housing, thus implying mechanically that policies reducing addictions and policies raising housing consumption are strong welfare complements even absent incentive effects. I am not sure there is an alternative model that would make more sense, but I would like the author to acknowledge and potentially provide more sensitivity to this modelling choice."*

[Conceded.] You and Referee 2, in his comment C5, identify the same problem, and it is more serious than a sensitivity issue: the complementarity between generosity and conditionality is the paper's headline result, and the specification was producing it mechanically.

You write that you are not sure there is an alternative that would make more sense. There is, and it is simply to remove the state dependence entirely. Preferences are now *fixed* across substance use disorder status: one set of weights, one subsistence parameter. Nothing about the utility function changes when a household acquires the disorder.

The argument for that is stronger than avoiding your mechanical complementarity, and it is worth stating because it also disciplines the rest of the paper. Any formulation that makes a household *more* susceptible to drugs by moving its preferences also makes it *better off*, mechanically, because it is now getting more utility out of something it was going to consume anyway. Flow utility is Cobb-Douglas in $b + v$, so the marginal value of the drug is proportional to $1/(b + v)$: a smaller subsistence parameter raises the marginal value of using, raises desired intake, and raises the level of utility with it. That sign is the wrong way round against everything known about substance use disorders, and it puts the welfare accounting the wrong way round too, since the policy that prevents a disorder would then be taking something valuable away.

The model therefore varies *income* rather than utility. What the disorder changes is the budget and the hazards: lost earnings, lost employment, higher mortality, harder recovery, each of them measured and each of them making the household worse off. The same rule governs what a child inherits, which matters because that is now the paper's central mechanism: a child inherits a worse productivity draw, not a stronger taste. Had it inherited the taste, its lifetime value would have risen, and the specification would be claiming that a child of a parent with a disorder is better off for it.

The obvious objection is that the state dependence was carrying a target, the gap between 44 per cent of income spent on drugs among people with a disorder and 11 per cent among experimenters. With fixed preferences that gap has to emerge endogenously, and it does, from forward looking restraint: a household without a disorder limits its use because use raises the probability of acquiring one, and a household that already has one faces no such margin. The gap becomes a prediction rather than an assumption, and the welfare complementarity becomes an incentive effect that the model has to produce rather than an artefact of the functional form.

The Associate Editor suggested an intermediate version, keeping the weights fixed but letting the subsistence parameter vary by status. I report why it does not work in the letter to him: the subsistence parameter shifts the level of drug spending, but the ratio between the two groups is governed by the curvature of the subsistence term, and opening a fourfold gap by moving it drives experimenters to a corner.

[Built since the first draft of this letter.] Done in the code, and taken further than I first proposed. The weights are identical clean and addicted, and so is the Stone-Geary shifter, so flow utility is the same function of consumption, housing and drugs in every state and after every choice. Becoming addicted relaxes no constraint and every private cost of the disorder comes from mortality, employment, the recovery hazard and the budget. My earlier draft kept a shifter that

varied with the disorder and with housing; I retired the housing dependence because housing is a choice, so a preference over it is a preference over a choice, and the dependence on the disorder because it is the same objection you raised about the weights, one step along.

M2(b) What patterns does the Stone Geary form actually generate?

> *"I find the use of the Stone-Geary functional form to generate a large mass of non-users, combined with income-dependent penalties . . . very clever. However, I am unclear on what are the actual patterns of drug consumption this generates, and where heterogeneity in v (conditional on addiction status d) stems from. It would be very helpful to present model-derived stylized facts on drug use . . . by income, employment status, housing status, assets . . . and explain how they depend on model parameters (notably b, the Stone-Geary parameter, and $\aleph$, the value of life)."*

Done, and it is now a section rather than a paragraph, because Referee 1's comment 1(d) and Referee 2's comment C4 ask for overlapping things and one exercise serves all three.

The section reports the drug use policy function by income and wealth, separately by disorder status, employment and housing, and then a channel shutdown ladder that re solves the model with one channel disabled at a time.

On $\aleph$ specifically, your question about how the patterns depend on it turned out to be more pointed than it looks. A structural check found that at the previous calibration, 64.6 per cent of the population had negative continuation value: they would rather be dead. For those households the mortality cost of drug use is a mortality *benefit*, which silently inverts the paper's central mechanism for exactly the population the policy targets. The reason was that $\aleph$ was calibrated to the income ratio among people with a disorder, which pins its *effect* on who uses but says nothing about its *level*. It is now pinned by the model implied value of a statistical life at 7.5 million dollars, which is what the paper always claimed to be doing, and the income ratio moves to the direct productivity penalty that the Associate Editor asks for in his comment A20.

M2(c) Where do the externalities come from if drug use is rational?

> *"Since drug use is 'rational' and agents are forward-looking, the model should make clearer where externalities arise that generate welfare-improving government interventions. There are potentially many sources of fiscal externalities . . . as well as potentially less obvious non-fiscal ones . . . It would be helpful to make those appear analytically or numerically, and establish how optimal policies change when some of this channels are shut down."*

This is the best question in the four reports and it now has its own section.

Analytically, the stylised model in Section 3 enumerates them, and one of them turns out to decide the paper.

The one that decides it is *intergenerational*, and it is the answer to your question rather than a fiscal side effect. A household that acquires the disorder, or that ends up on the street, passes a worse productivity draw to the child who replaces it, and counts only a fraction ω of that loss. The complement is external in the strict sense you mean: it is a cost the decision maker does not bear and cannot be made to bear by any price it faces. It is also why rational forward looking behaviour is not enough to make the supports unnecessary. Every household in the model is optimising, and the ranking of the two policies still reverses once the children are counted. Section 3.6 proves that in the stylised model: without the externality Housing First weakly dominates, and with it the dominance goes.

Two more are fiscal: a household that becomes homeless imposes a service cost on the budget, and a household that acquires a disorder lowers its own expected tax payments through employment and mortality. One further one is not fiscal: following Referee 2's comment C2, a disutility from the size of the homeless population enters period utility, and no individual household internalises its own contribution to that stock. That one is in the written model and not yet in the code, which I flag in the letter to Referee 2 as well.

Numerically, the channel shutdown ladder re solves the model with each externality switched off and reports how the optimal policy moves, which is exactly the exercise you propose.

[New result.] There is also a third exercise, which came directly out of thinking about your comment, and I think it is the paper's most interesting single table. Taken to its conclusion, the position implicit in your question is that households internalise their own costs and the supports are therefore unnecessary. That is a *testable claim about the existing safety net*, not an abstract point about rationality, because the benchmark economy already insures homelessness through vouchers, food stamps and disability. So I test it: remove the net, and then remove it again with the drug choice frozen at its benchmark schedule.

The two removals differ by the *addiction margin*. The classic moral hazard from social insurance is consumption smoothing, and one bad year is one bad year. Here there is a second margin, because cushioning the downside of homelessness also cushions the downside of acquiring a disorder, which is the state that leads there. That margin is worse than the savings one for a specific reason: it is persistent and self reinforcing. A one period distortion becomes a trap. The gap between the two arms measures how much of the moral hazard runs through addiction rather than through ordinary consumption smoothing, and the sign of the welfare number answers your question directly.

M2(d) How is the price elasticity targeted when p_v is normalised to one?

> *"I am somewhat unclear how the elasticity of drug use to its price is targeted, given that the price of drugs p_v is normalized to 1. The author says they vary the price of drugs in the model, but this requires suppressing this normalization and potentially affects the definition of other prices in the model?"*

You are right that one cannot simultaneously normalise a price and vary it, and the previous draft did not say clearly enough which it was doing.

The drug price is now an equilibrium object that clears a supply curve, which Referee 1's comment 1(c) and the Associate Editor also ask for. The normalisation is stated explicitly and confined to where it belongs: the drug price is the numeraire of the *first stage* of the calibration, held at one there, and a scale parameter of the supply curve is renormalised at each market clearing so that the benchmark allocation is invariant to the supply elasticity. The price is then free in every counterfactual and in the elasticity computation, so nothing is both fixed and varied.

Because the supply elasticity now does real work, I discipline it rather than assume it. There is no direct estimate for an illicit drug, and cannabis is the closest market to one: it remains a Schedule I substance federally, so even a state licensed producer works without interstate trade, without ordinary banking and under federal prohibition, which is the set of frictions an illicit supplier faces. Mace, Patel and Seegert (2020) estimate a price elasticity of supply of 1.46 on Washington State administrative data, and that is the number the model uses. I state the caveat that it is a legal licensed market with a capped canopy. I do not bracket it with cigarettes or gasoline, which was the previous draft's approach: those markets share none of the frictions above, so their elasticities are not informative about this one.

I also explain why the frequently quoted RAND figure of minus 3.6 is not the right object here. A negative supply elasticity reflects enforcement swamping, and Caulkins and Reuter note that swamping "pertains to total drug volume . . . not just or primarily drug specific market volume", so it operates on the aggregate market rather than at my margin.

[Built since the first draft of this letter.] The supply curve is finite rather than flat, and it now clears in the code rather than only in the written model. The elasticity is 1.46, from Mace, Patel and Seegert (2020) on Washington State administrative data. The price is one at the benchmark by the normalisation of the supply shifter, so the calibration itself never has to clear the drug market; the curve matters where quantity moves away from the benchmark, which is the counterfactuals, and that is where it operates.

Minor concerns

m1 The labour supply elasticity should matter more

> *"I am surprised that there is not a clearer role for the labor supply elasticity in welfare calculations, as it should matter quite substantially for the welfare costs of balancing the budget with extra labor taxes."*

Agreed, and I flag for you that the Associate Editor's comment A3 pushes the other way, asking what role labour supply plays in the framework at all. I have kept it, on your reasoning: every counterfactual in the paper is financed by the labour income tax, so the elasticity is not decorative.

The reason its role was invisible is that the previous decomposition did not separate the mechanical transfer from the deadweight loss. It does now, by re solving the counterfactual with the hours choice frozen at its benchmark schedule. Freezing the choice set preserves the Bellman identity; the previous approach, of putting a large penalty on the disutility of labour, drove hours to a corner rather than holding them fixed, which is a different experiment and gave a misleading answer. The deadweight column is now a clean number and the reader can judge for themselves.

m2 Present sensitivity as graphs rather than tables

> *"Some of the results of the counter-factual experiments might be more helpfully presented in the form of graphs comparing outcomes and sensitivity to parameters, rather than tables, for example, for the sensitivity to the loss of consortium spillover, or to the value-of-life parameter* ℵ*."*

Done for ℵ and for the other parameters where sensitivity is reported.

The loss of consortium case is moot: that parameter is deleted. Referee 1's comments 3(a) and 3(b) argued that an adjustment applied after the model is solved is internally inconsistent and that the conservative value of a parameter chosen without a source is zero. I agree, and the externality inside period utility from Referee 2's comment C2 does the job properly instead.

m3 Vary the voucher receipt probability continuously

> *"It would be helpful to provide a more continuous experiment that varies the probability of voucher receipt rather than go straight for full rationing to fully universal provision. This would help establish whether there is a Laffer curve of voucher generosity, or whether the corner solution of universal provision is indeed optimal, and what are the key parameters governing this."*

Done, and it has become one of the paper's main figures. The probability of voucher receipt is swept from the benchmark to universal, with welfare, homelessness, drug use, substance use disorder and the labour income tax plotted along it. Referee 2's comment C6 asks a related question about broadening eligibility, and the same figure answers it.

I run the sweep twice, with and without the sobriety condition. That turns the paper's central claim into the vertical gap between two curves, and makes visible something four table cells could not: whether the gap *widens* with generosity. It should, because conditionality is worth more when the benefit being conditioned on is guaranteed than when it is rationed, and that cross partial is what the stylised model in Section 3 proves. Presenting it this way also softens the headline in the way Referee 1's comment 3(c) asks for, since the claim becomes one about an interaction rather than about a winner.